Mass Spectrometry

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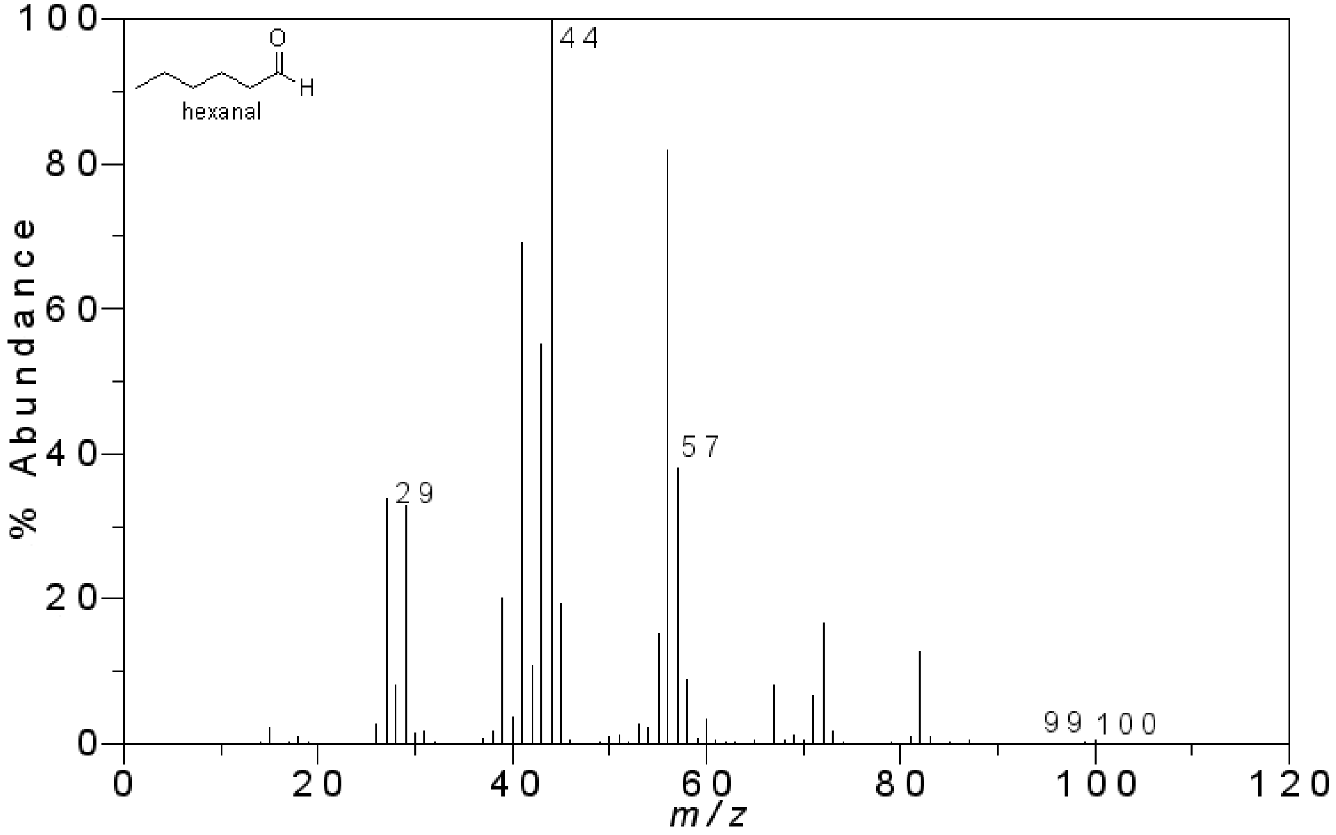



POWER OFF ELECTRONICS BEFORE REMOVING SOURCE
Thermo
LTQ XL
Thermo

- Most sensitive technique for structural elucidation (determination)

- This technique is capable of providing information about:
  - Elemental composition of samples of matter
  - The structures of organic, inorganic and biological molecules
  - Qualitative and quantitative composition of complex mixtures
  - The structure and composition of solid surfaces
  - Isotopic ratio of atoms in a sample

- Samples are converted into rapidly moving positive ions and ion fragments
- Ions are then separated and characterized
- The study of charged molecules and fragment ions produced from a sample exposed to ionizing conditions
- The study of relative intensity spectrum
- Results from the correlation of ions with their mass to charge ratio (m/e)
- Largest peaks is arbitrarily assigned a value of **100** and is termed as **base peak** (100% peak)
- Heights of the remaining peaks are computed as percentage of base-peak height

Mechanism of cation production

- Most common sources are:

Gaseous sources (For gaseous or volatile samples)

- Electron impact ionization
- Chemical ionization
- Field ionization
- **Desorption sources** (For non-volatile liquids or solids)
  - Field desorption
  - Thermal ionization (Surface ionization)
  - Fast atom bombardment (FAB)
  - Photoionization
  - LASER Desorption
  - Matrix assisted LASER desorption ionization (MALDI)

- Electron Spray Ionization (ESI)

- Depending upon the energy of ionization, method may be **hard** or **soft**

Principle

- Charged molecules or molecular fragments are generated in a high-vacuum region, or immediately prior to a sample entering a high-vacuum region, using a variety of methods for ion production.

- The ions are generated in the gas phase so that they can then be manipulated by the application of either electric or magnetic fields to enable the determination of their molecular weights.

INSTRUMENTATION

- IONISATION SOURCE
- ANALYSER
- DETECTOR
- VACUUM SYSTEM

IONISATION SOURCE → ANALYSER → DETECTOR → RECORDER

1. IONISATION SOURCE

- Ionization of sample occurs in ionization chamber.

The minimum energy required to ionize an atom or molecule is referred to as “Ionization potential”

- The appearance of mass spectrum for a given molecular species is highly dependent upon the method used for ion formation.

Ionization sources include

- GAS PHASE SOURCE
- DESORPTION SOURCES

GAS PHASE SOURCE (For gaseous or volatile samples)

Volatilization may be carried out externally (batch inlet system), or internally from a heated probe.

- Electron impact ionization
- Chemical ionization
- Field ionization

- **Desorption Source**

bulk sample vaporization (For non-volatile liquids or solids)

  - Field desorption
  - Thermal ionization (Surface ionization)
  - Fast atom bombardment (FAB)
  - Photoionization
  - LASER Desorption
  - Matrix assisted LASER desorption ionization (MALDI)
  - Electron Spray Ionization

A. Gaseous Source

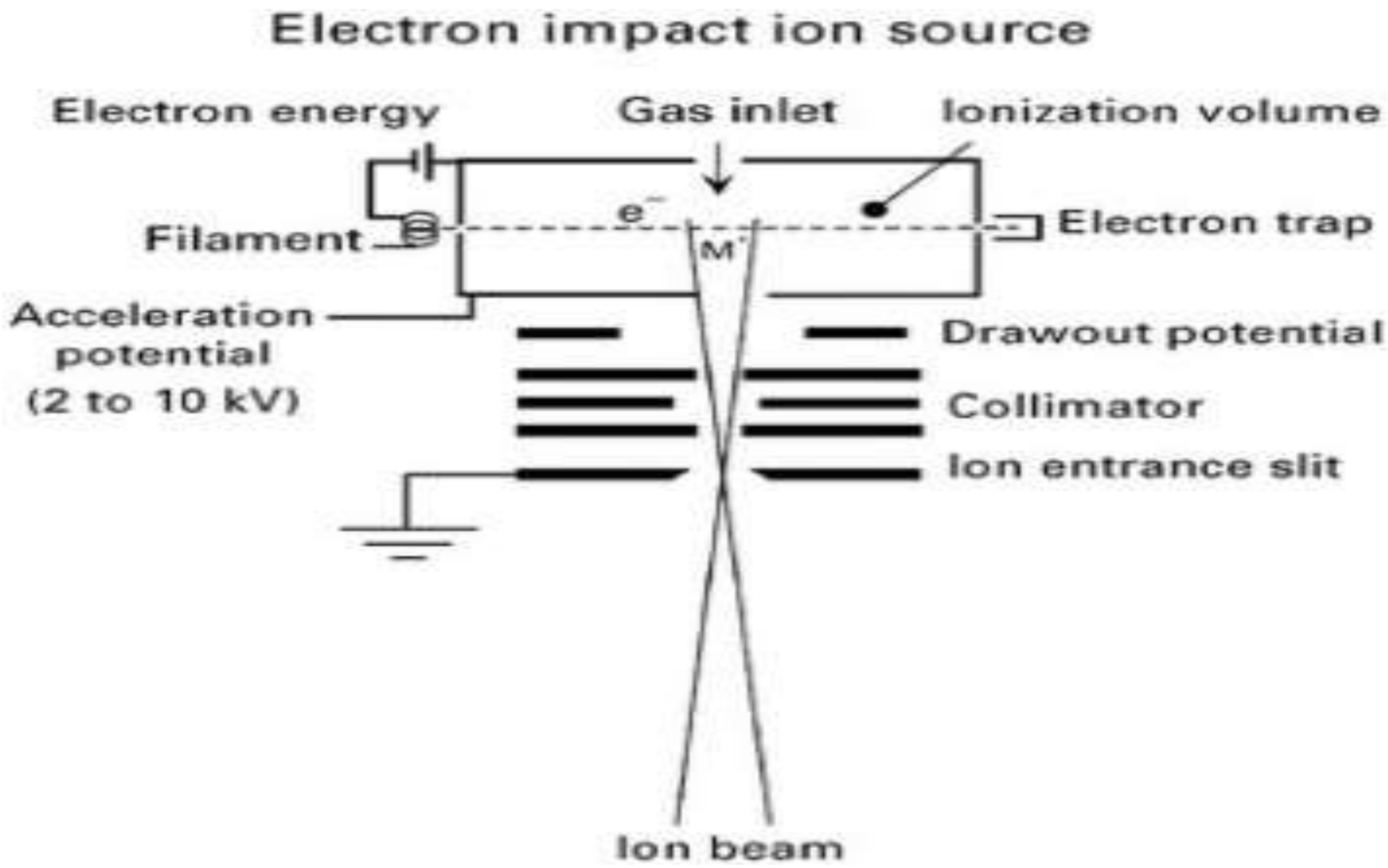
- Sample inserted in the ionization chamber in vapor form .

1. Electron impact Ionization:

- Sample molecules in vaporized state are bombarded with energetic electrons from hot rhenium or tungsten filament.
- These electrons are collimated by an electric field
- The sample pressure is of order 10^{-5} to 10^{-6} torr.
- The vapor of the sample to be analyzed is introduced at right angles to the electron beam.
- Energy transfer takes place between the bombarding electrons and the neutral molecules.

- When the transferred energy is equal to the ionization potential of the molecule, it loses an electron and ionizes.
- For most elements, the ionization potential is in the range of 5 to 15eV (it varies from 8 to 12eV) for organic compounds.
- When more energy is transferred to the molecule by the bombarding electron, a hypothetical molecule, molecule A-B can undergo the various type of electron impact induced reactions
- Generally, the spectrum is run at 70eV (average energy of electron beam)

- Electrical potential of bombarding electrons just highs enough to initiate fragmentation is called **appearance potential**
- The magnitude of appearance potential is equal to or greater than the sum of dissociation energy of the fragmented bond and ionization
- Bombarding electrons have a high energy
- More than one bond in a molecule can be broken



Schematic diagram of an electron impact ion source

The primary product is singly charged positive ion (cation) only one electron is abstracted in this method.

$A—B + e \rightarrow A—B^*$ (Excitation)

$A—B + e \rightarrow A—B^{+\bullet} + 2e$ (Single ionization)

$A—B + e \rightarrow A—B^{n+1} + (n+1)e$ (Multiple ionization)

$A—B + e \rightarrow A^+ B^- + e$ (Ion pair production)

$A—B + e \rightarrow A^+ + B + 2e$ (Dissociative ionization)

$A—B + e \rightarrow A—B^-$ (Electron capture)

- Due to high energy, ions of lower mass are formed , Such ions are called **daughter ions**

2. Chemical Ionization

- Useful technique for organic compounds.
- A reaction gas (e.g hydrocarbon CH_4, iso butane, ammonia, nitric oxide, tetramethylsilane) is introduced into the ionization chamber of mass spectrometer at a pressure of 1 torr.
- Primary ionization of the reaction gas is effected by electron impact (50 to 70 eV)
- Because of the high pressure, these primary ions undergo ion-molecule reaction with the neutral molecules of the reaction gas and with the sample molecules which are present at relatively low concentration (-1%)

- If the reaction gas is methane, it react with high energy electron to give primary ions e.g., CH_4^+, CH_3^+, CH_2^+.
- These ions react rapidly with additional methane molecules as:

$$CH_4^+ + CH_4 \rightarrow CH_5^+ + CH_3^-$$

$$CH_3^+ + CH_4 \rightarrow C_2H_5^+ + H_2$$

- Collision between sample molecule (XH) and highly reactive CH_5^+ or $C_2H_5^+$ occurs proton or hydride may transfer

$$CH_5^+ + XH \rightarrow XH_2^+ + CH_4 \quad \text{(Proton transfer)}$$

$$C_2H_5^+ + XH \rightarrow XH_2^+ + C_2H_4 \ \text{(Proton transfer)}$$

$$C_2H_5^+ + XH \rightarrow X^+ + C_2H_6 \ \text{(Hydride transfer)}$$

- Proton transfer gives $(M+1)^+$ ions
- Hydride transfer gives $(M-1)^+$ ions
- Chemical ionization is useful for organic compounds
- Spectrum formed is simple and supplement to electron impact

Field ionization

- Ions are formed due to very strong electric field
    - Electric field strength is 10^8 V/cm
    - Voltage is 10 to 20 kV (high voltage)
- Specially formed emitters are used
    - Fine tips having diameter of less than 1µm diameter
    - Sample is diffused into the ionization chamber
- Molecular ion peak is significant ions peak
- Fragmentation process is minimum
    - Less number of peaks
    - Simple (non-complex) spectrum

B. Desorption sources of ionization

- Desorption is the process in which atomic or molecular species leave the surface of a solid and escape into surroundings
- These methods are used for non-volatile substances, delicate biomedical species with molar mass 10^3 Da
- Energy in various forms is transferred into the solid or liquid sample, cause direct formation of gaseous ions

Field desorption

- High potential electrode is used
- Electrode is mounted on a removable probe
- Sample is applied on this electrode and probe is inserted into the sample compartment
- Ionization takes place by high potential
- Gives simpler spectra

Thermal ionization (surface ionization)

- Sample is applied on a filament wire or ribbon and inserted into the ion source
- Part of the substance on hot filament is vaporized in the vacuum
- Some of the atoms or molecules are vaporized directly as ions.
- Only useful for substances with low IP.
- Highly sensitive technique

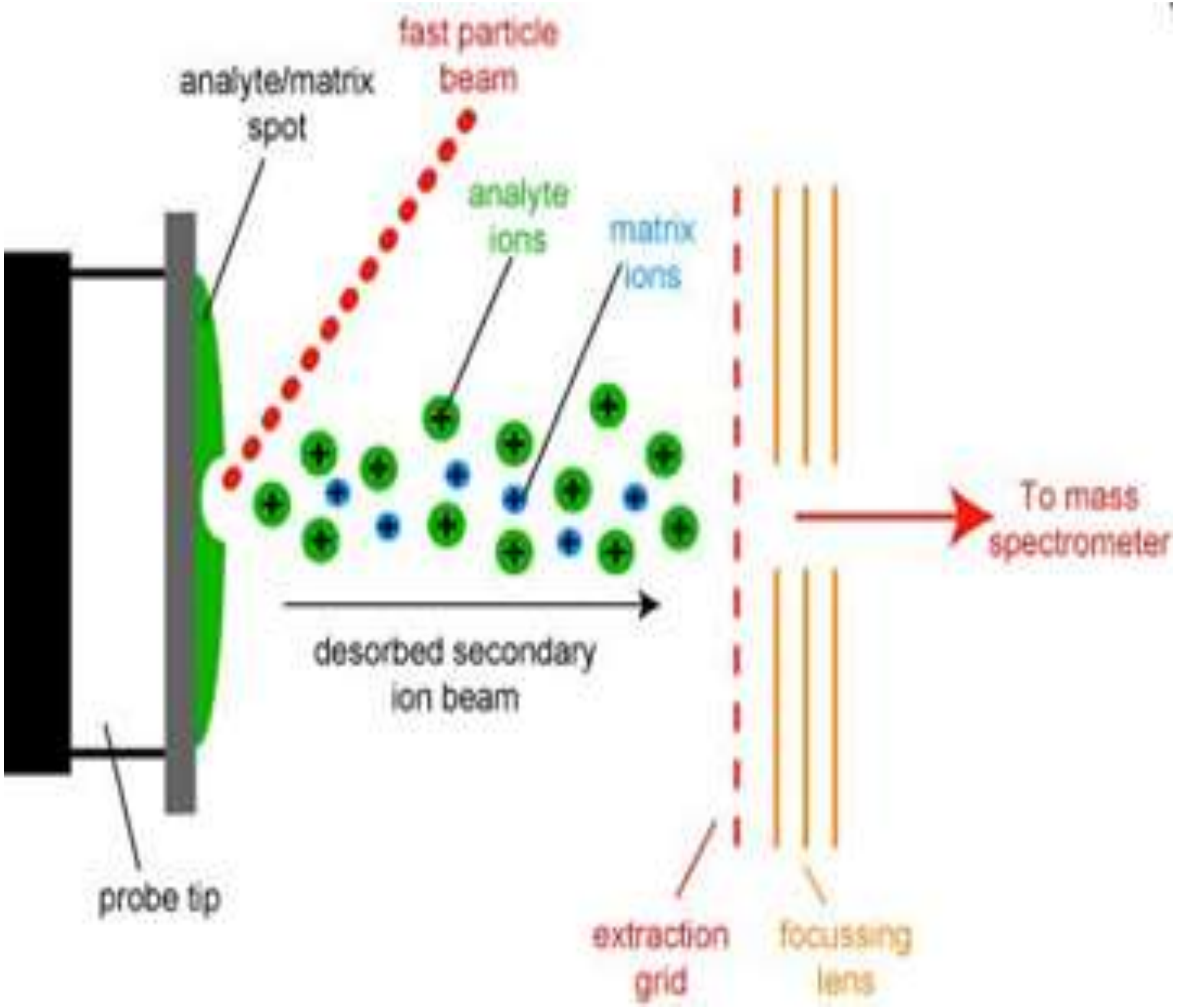
Fast Atom bombardment ionization

- Sample is condensed in a matrix placed on a sample probe
- Bombarded with high energy beam of xenon or argon atoms
- Sample and matrix are sputtered from the probe surface
- Matrix is used to dissolve the sample and facilitate desorption as well as ionization

  e.g of MATRIX USED .

  - glycerol matrix
  - 3-nitrobenzyl alcohol matrix
- Beam of fast atoms is obtained by passing accelerated argon or xenon ions from an ion source or gun through a chamber containing argon or xenon atoms at a low pressure (10^{-5} torr)

- Speeding ions undergo a resonant electron exchange reaction with atoms without substantial loss of translational energy.
- With FAB technology molecular weight over 10’000 can be determined
- Detailed structure can be obtained
- Spectrum also contains peak due to matrix


fast particle beam
analyte/matrix spot
analyte ions
matrix ions
To mass spectrometer
desorbed secondary ion beam
probe tip
extraction grid
focussing lens

Photoionization

- Ionization of molecules can be accomplished by a beam of UV light of sufficiently short wavelength.
- Many ionization processes occur at 10 eV or more.
- This corresponds to 1200 A or less
- Ionization occurs by collision between photons and neutral molecules.

$$A + \quad hv \rightarrow A+ \quad e$$

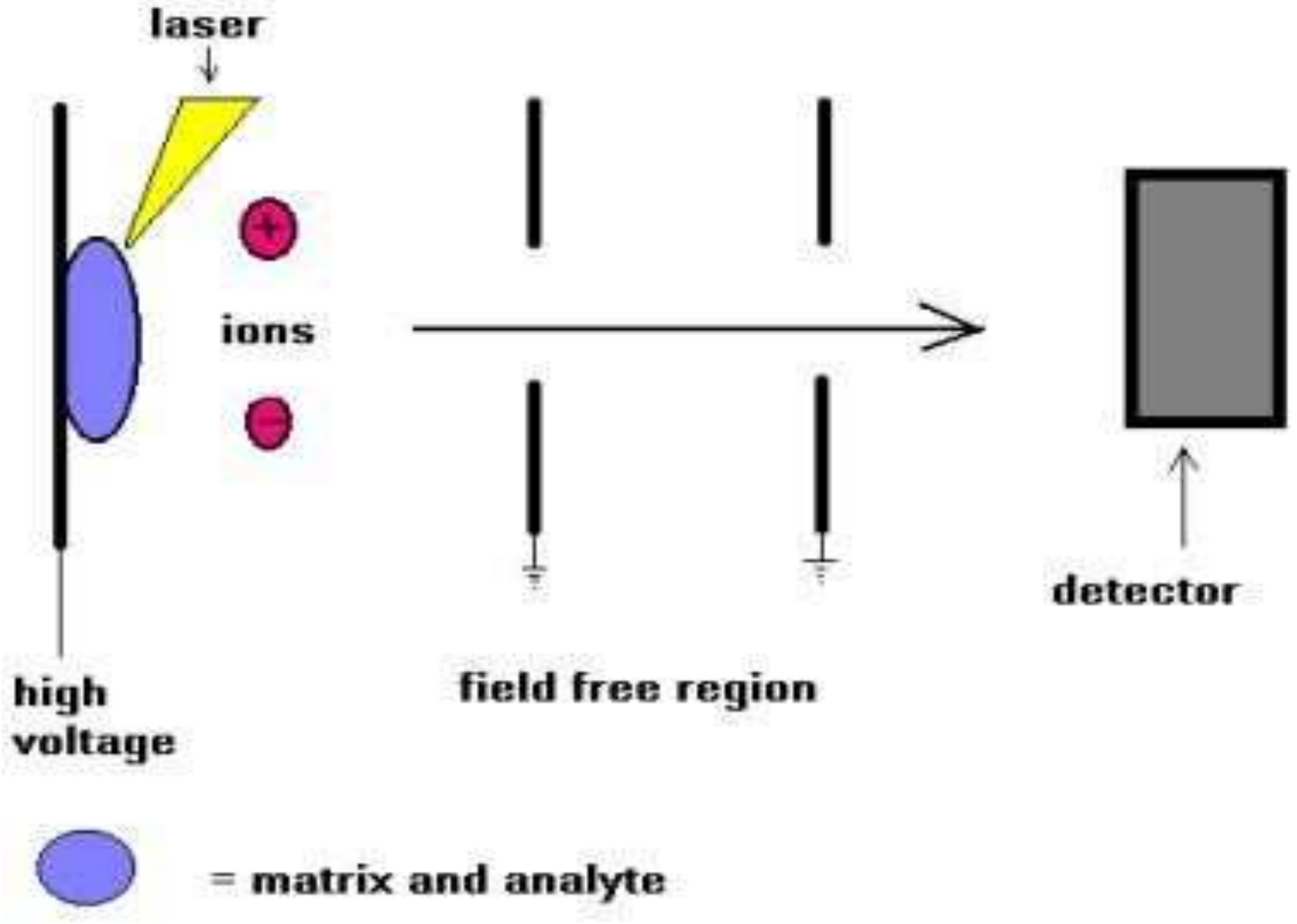
Laser Desoprtion

- Laser beam also used for ionization

Matrix assisted laser desorption Ionisation (MALDI)

- Based on bombardment of sample molecules in matrix with a laser light to bring out the ionization.
- The sample is pre-mixed with highly absorbing matrix compound (nicotinic acid, sinapinic acid, urea and derivatives) for most consistent and reliable results.

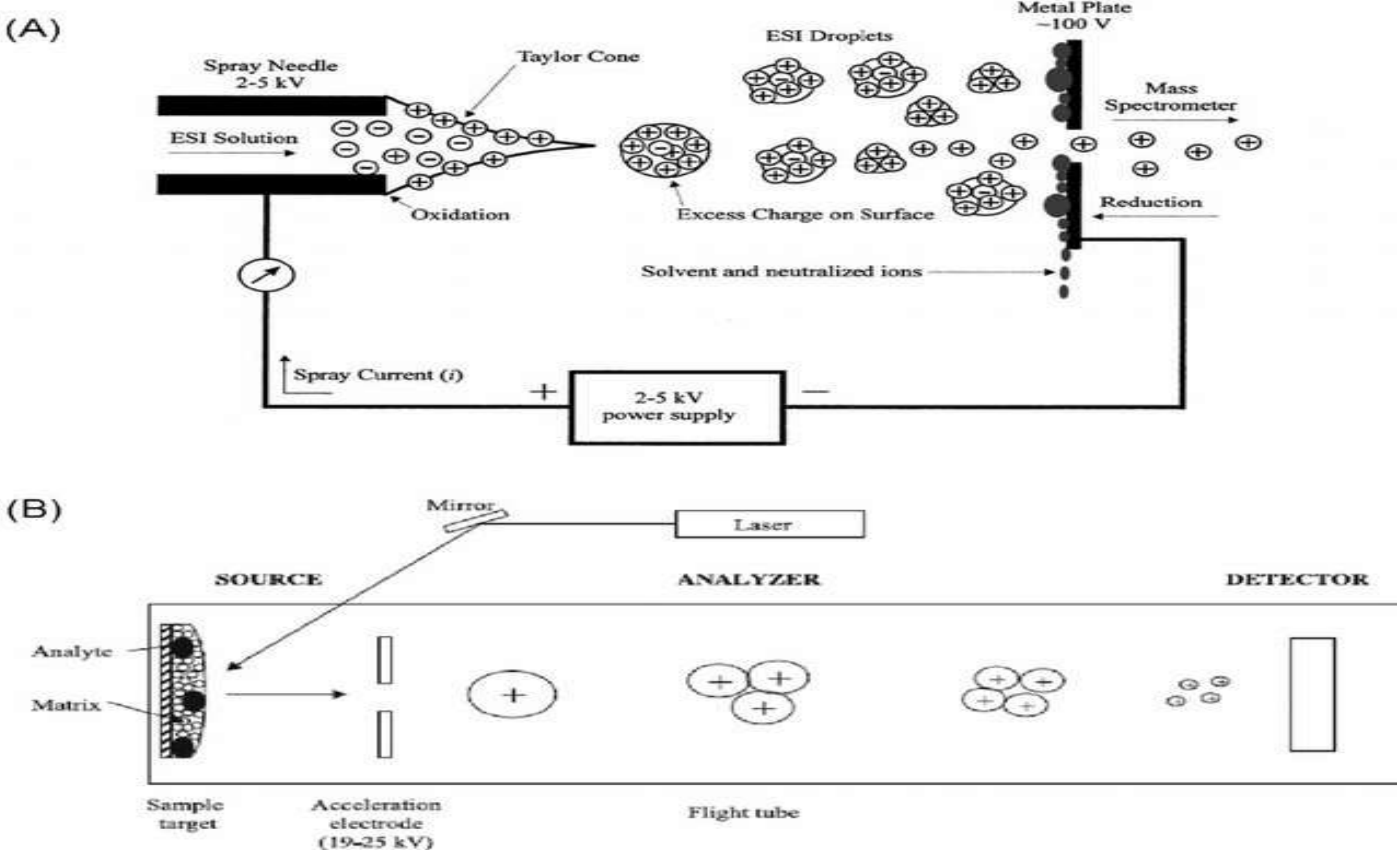
- The laser is fired and energy arriving at the sample/matrix surface optimized.
- The matrix transforms the laser energy into excitation energy for the sample for causing ionization.
- Energy transfer is efficient and also the analyte molecules are spared excessive direct energy that may otherwise cause decomposition.


laser
ions
field free region
detector
high
voltage
= matrix and analyte

Electron Spray Ionisation (ESI)

- One of the atmospheric pressure Ionization (API) technique and well suited for the analysis of polar molecules.
- Sample is dissolved in polar volatile solvent and pumped through a narrow, stainless steel capillary (75- 150 um) at a flow rate of 1ul/minutes to form droplets.
- droplet carry charge when exit the capillary and as the solvent vaporizes, the droplet disappears leaving highly charged sample molecules.
- A high voltage of 3 or 4 kV is applied to the tip of capillary, which is situated within the ionization source of mass spectrometer and a consequence of this strong electric field, the sample emerging from the tip is dispersed into an aerosol of highly charged droplets.

- The process is aided by a co-axially introduced nebulizing gas flow around the outside of the capillary.
- For this technique, sample must be soluble in low boiling solvents (acetonitrile,CHCl3 etc)
- Stable at a very low concentration, i.e. 10^{-2} mole/L.


(A)
Spray Needle
2-5 kV
Taylor Cone
ESI Droplets
Metal Plate
~100 V
Mass
Spectrometer
ESI Solution
Oxidation
Excess Charge on Surface
Reduction
Solvent and neutralized ions
Spray Current (i)
+
2-5 kV
power supply
−
(B)
Mirror
Laser
SOURCE
ANALYZER
DETECTOR
Analyte
Matrix
Sample
target
Acceleration
electrode
(19-25 kV)
Flight tube

2. MASS ANALYSER

- Ions produced are accelerated by accelerated potential (2-8 kV).
- These ions enter mass analyzer which differentiate them on basis of mass to charge ratio by applying magnetic and electric field.

- Two main functions of a mass analyzer

a) Resolve the ion beam into its component ions
b) To maximize the resolved ions intensities (dispersive focusing)

- Electrostatic analyzer focuses or directs the ions towards magnetic analyzer
- Also called ion separator

Consists of:

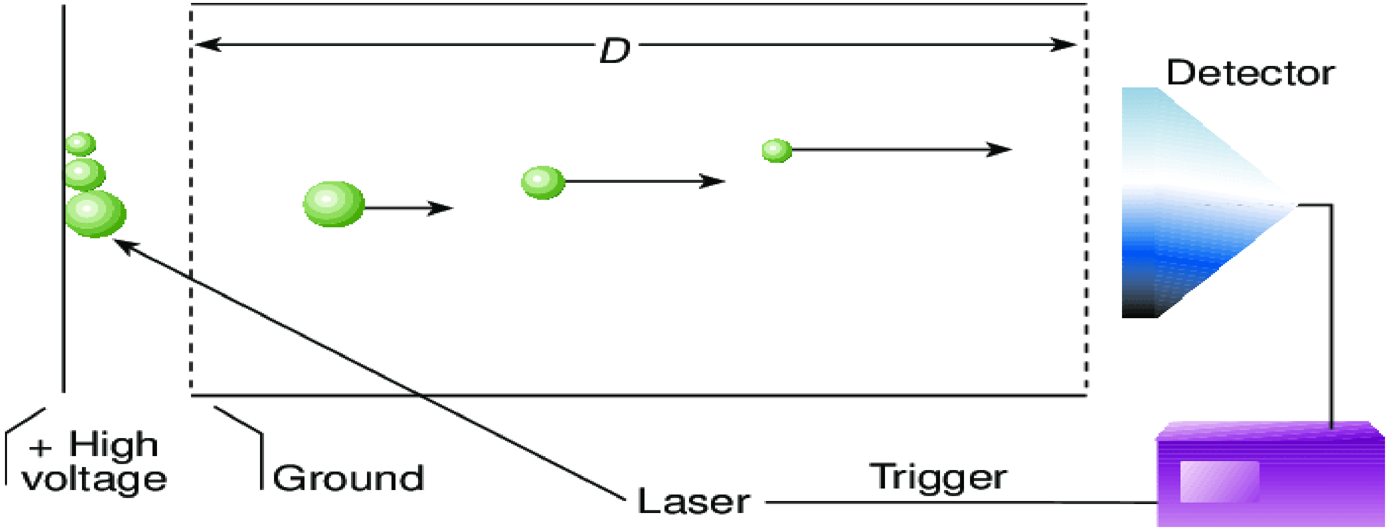
1. Magnetic sector analyzer
2. Time of flight analyzer
3. Quadruple mass analyzer
4. Ion trap analysers

Magnetic sector analyzer

- Two types of magnetic sector analyzer
  - Single focusing analyzer
  - Double focusing analyzer
- Single focusing analyzer
  - Ions are separated only by magnetic field
- Double focusing analyzer
  - Electrostatic analyzer is used in addition to magnetic one

Time of Flight (ToF) analyser

- Measures the time required for an ion to travel from the ion source to detector
- Lighter particles arrive at the detector earlier than the heavier ones.
- Typical flight times are 1 to 30 us
- Resolution power is less than 1000 and also have low mass ranges



Ion trap analysers

- An analytic ion from an electron impact or chemical ionization source is entered in it.
- Ions an appropriate m/e value circulate in the orbit.
- As the radio frequency voltage is increased, the orbits of heavier ions become stabilized while those of lighter ions become destabilized (trapped) goes to the detector (leave the chamber)

The quadrupole mass analyser

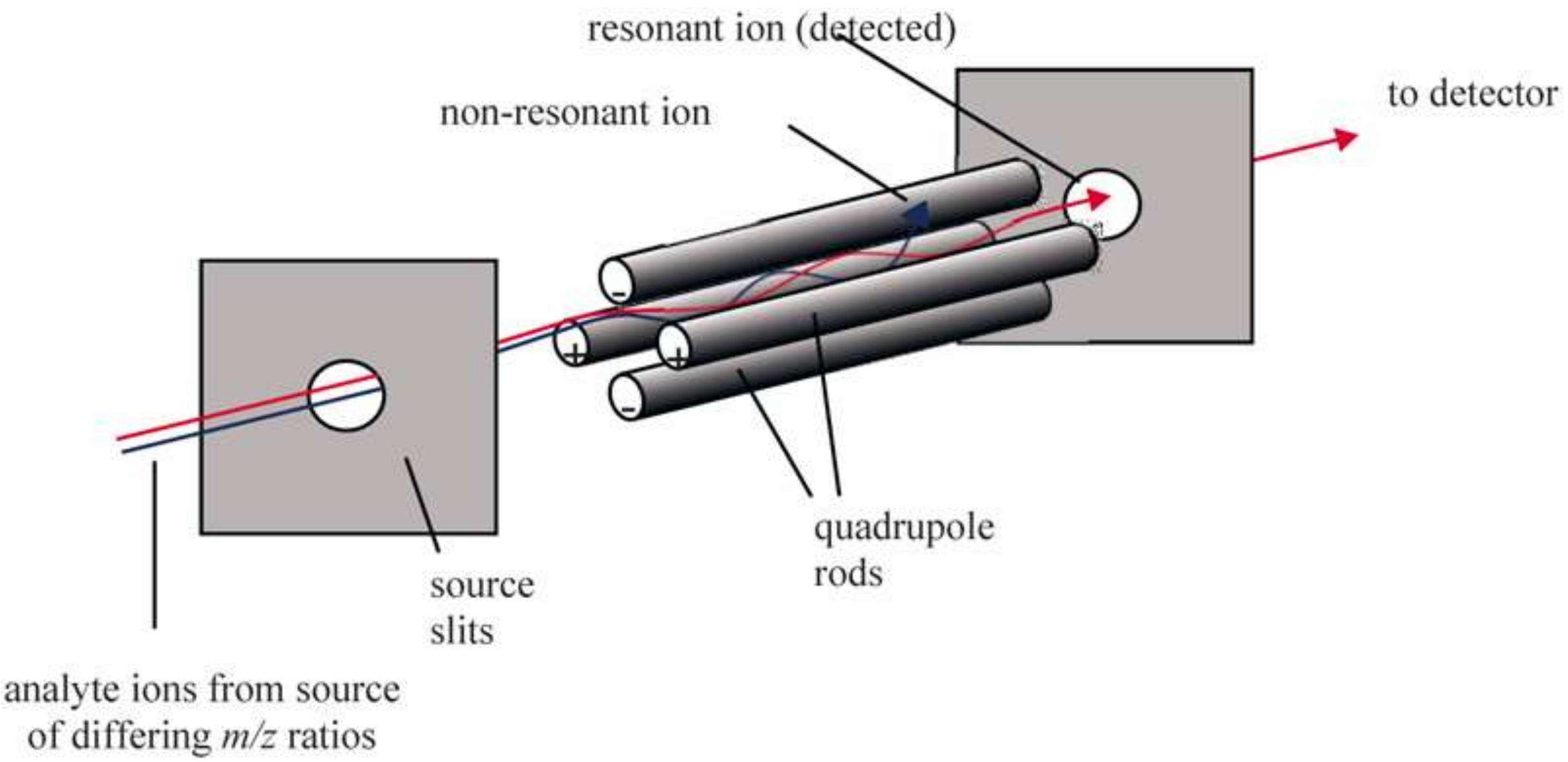
The principle of quadrapole mass analyser is that the trajectory of a particle moving through the lines of force of two fields at continuous current is modulated by a radio frequency.

Has 4 metal bars, fixed at angles of squares and ceremically isolated, connected alternatively to form two couples to which are applied DC and RF potential with charges of opposite signs.

So, the accelerated particles enter the analyser and begins to oscillate in a complex manner according to their m/e and applied RF/DC ration.

- For every value of these ratios, only one mass is able to pass completely through the filter and impringe on the collector.
- The other ions of the other m/e value strikes the pole and discharge.
- Therefor, by varying the RF/DC ratio, it is possible to select a single value of m/e able to give a signal.
- By continuous variation from minimum to maximum, the whole spectrum can be recorded

Quadrupole mass analyser



3. Detectors

3 methods of ions detection

i. The electron multiplier
ii. Photographic plates
iii. Scintillation type detector

The electron multiplier

- Ions strike on photo emissive surface (present to cathode) to release the electron.
- The electrons are attracted towards the anode to generate the current which is measured.
- The cathode and several dynodes have Cu/Be surfaces
- The intensity of current is proportional to number of electrons (no. of ions reach to detector).
- Strength of current is measured and recorded.

Photographic plates

- Plates are coated with a *silver bromide emulsion* are sensitive to energetic ions.
- All these ions are allowed to fall on the photographic plates, to cause impressions.
- This detector is well suited for wide range of m/e values.

Scintillation type detector

- Crystalline phosphor dispersed on thin aluminum sheet that is mounted on photomultiplier tube.
- The ions strike the scintillator to emit out the light which is detected by photomultiplier tube.

4. Vacuum System

- The vacuum system10-7 to 10-8 torr is required to avoid sputtering of ions.
- This is produced with mechanical attached with diffusion pump.

Amplifier and Read Out system

- Secondary electrons generated by detectors are further amplified by amplifier
- Amplified current is recorded
- The pen and ink type recorders are slow for mass spectrometers work
- Common arrangement involves the use of 3 to 5 galvanometer mirrors which deflect UV beam of UV radiation into a UV sensitive recording paper as ions are collected on the electrometer.
- Microprocessor bases MS with computer printer plotter attachments are used now days

Applications of Mass Spectrometer

- Most sensitive technique for qualitative analysis
- Identification of isotopes
  - Percentage abundance of isotopes
- Study of free radicals

- Calculation of molecular formula
- Analysis of closely related compounds
    - Hydrocarbons, petroleum products and lubricant oils
- Structure of Biomolecules
    - Carbohydrates, proteins, nucleic acids
- Determination of molecular weight
- Inorganic trace analysis of elements
    - In alloys and in super conductors
- Quantitative analysis of complex organic mixtures

General Applications

- To identify environmental pollutants
  - E.g. water pollution by dioxins
  - Industrial waste incineration product (dioxins) in fishes
- Composition of molecular species found in space
- Illegal use of steroids by athletes
- Adulteration of honey with corn syrup
- Locate oil deposits in rocks(petroleum precursor)
- Monitor fermentation process in biotech industry
- Elemental composition of semiconductor material
- Sequence of biopolymers

    - Proteins, oligosaccharides
- Identify drug metabolism products
- Forensic analysis for drugs abuse
- Age & origin of specimens in geo-chemistry and archaeology